The Tate Conjecture

Notes

César Galindo

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Abstract

These notes present a structural exposition of the Tate conjecture with emphasis on the algebro-geometric mechanisms that underlie its various formulations. The guiding point is that the conjecture should not be read merely as a statement about fixed subspaces in ℓ-adic cohomology, but as a comparison between the geometry of cycles and correspondences and the linear algebra produced by cohomological realizations with Galois action. Accordingly, particular weight is placed on absolute purity, refined Gysin morphisms, the passage from divisors to the Picard scheme and the Néron–Severi lattice, the role of numerical equivalence in the semisimplicity of motives, the algebraicity of K"unneth projectors, and the special geometry that makes abelian varieties and K3 surfaces accessible. The later sections place the conjecture in a broader motivic, Hodge-theoretic, and Tannakian context, with special attention to the precise points at which full faithfulness, independence of ℓ, and the standard conjectures enter.

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1. Introduction

The Tate conjecture belongs to the small group of statements that reorganize an entire area. It is easy to formulate, but its meaning is distributed across several layers of geometry. At the most concrete level it concerns codimension-r algebraic cycles on a smooth projective variety and their images in ℓ-adic cohomology. At a deeper level it is a statement about correspondences, because the relevant cohomology groups are not only receptacles for classes but also targets for algebraic endomorphisms induced by cycles on self-products. At a still deeper level it is a statement about the extent to which realization theory remembers the algebraic structure of a variety.

Fix a smooth projective variety X over a finitely generated field k and a prime $\ell \neq \mathrm{char}(k)$. The conjecture predicts that the G_k-invariant part of $H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))$ is exhausted by classes of algebraic cycles. Over a finite field this can be restated in terms of the q^r-eigenspace of Frobenius; over a number field it becomes a statement about global Galois invariants inside a family of local comparison theories. In either case the point is the same: the symmetry detected by arithmetic should already come from geometry.

A first source of richness is that the conjecture naturally sits between two comparison problems. The first concerns equivalence relations on cycles:

$$\sim_{\text{rat}} \subseteq \sim_{\text{alg}} \subseteq \sim_{\text{hom},\ell} \subseteq \sim_{\text{num}} .$$

The second concerns realization functors: Betti, de Rham, crystalline, and ℓ-adic realizations all retain different aspects of the same underlying geometry. The Tate conjecture lies precisely at the interface of these two comparison problems. It asks whether the Galois-theoretically distinguished part of a realization is already visible on the algebraic side.

For this reason the conjecture is best understood not as an isolated assertion but as part of a larger architecture. On the geometric side one has Chow groups, correspondences, Picard and Albanese type constructions, and eventually the category of pure motives. On the arithmetic side one has Frobenius operators, semisimplicity phenomena, and monodromy groups. The connection between the two is mediated by cycle class maps and the action of correspondences on cohomology.

Plan of the exposition. We begin by recalling the cycle-theoretic input with enough precision to make the later motivic reformulations meaningful: local cycle classes, Gysin morphisms, and the hierarchy of equivalence relations. We then pass to the Frobenius description over finite fields and to the divisor case, where the Picard scheme and the Néron–Severi lattice already reveal the basic mechanism of the conjecture. The first decisive theorem is Tate's theorem on endomorphisms of abelian varieties over finite fields, followed by the characteristic-zero analogue supplied by Faltings. From there we explain why the conjecture becomes naturally motivic, how Künneth projectors and the standard conjectures intervene, and why K3 surfaces require a genuinely different package involving crystalline theory and Kuga–Satake. The final sections return to the broader picture: integral refinements, Brauer obstructions, Tannakian symmetry, and the comparison with Hodge classes.

The guiding diagram is the following.

$$\begin{tikzcd} \mathrm{CH}^r(X)_{\mathbb{Q}} \arrow[r, "\mathrm{cl}_\ell"] \arrow[d] & H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))^{G_k} \\ \mathrm{Corr}^0(X,X) \arrow[r] & \mathrm{End}_{G_k}\big(H^\bullet_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell)\big) \arrow[u] \end{tikzcd}$$

The vertical arrows should be read schematically. A cycle on X gives a class; a cycle on $X \times X$ gives an endomorphism of cohomology. The conjecture asserts that the arithmetic symmetry visible on the right should ultimately be accounted for by algebraic correspondences on the left. This is why the Tate conjecture is so closely linked to the standard conjectures and to the expectation that realizations of motives should be fully faithful.

2. Foundations: cycle classes and Weil cohomology

2.1. Geometric setup and ℓ-adic cohomology

Let k be a field, let $\bar{k}/k$ be a separable closure, and write $G_k = \mathrm{Gal}(\bar{k}/k)$. For a smooth projective variety X/k and a prime $\ell \neq \mathrm{char}(k)$, one defines

$$H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Z}_\ell) := \varprojlim_n H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Z}/\ell^n\mathbb{Z}), \qquad H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell) := H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

These are finite-dimensional $\mathbb{Q}_\ell$-vector spaces endowed with a continuous G_k-action. For the purposes of the Tate conjecture, however, it is important to remember slightly more than this bare definition. One needs a theory with supports, refined Gysin morphisms, and a satisfactory formalism for cup-products and correspondences. In practice this extra structure is supplied by absolute purity and the six-functor formalism.

If $i\colon Z \hookrightarrow X$ is a regular closed immersion of codimension r, absolute purity identifies the cohomology with supports along Z with the shifted and twisted cohomology of Z:

$$H^{m-2r}_{\text{ét}}(Z_{\bar{k}}, \mathbb{Q}_\ell(-r)) \;\cong\; H^m_{Z,\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell).$$

In degree $m = 2r$ this produces the local fundamental class of Z. The global cycle class map is assembled from these local classes together with additivity and passage to rational equivalence.

The formal properties used throughout are finite-dimensionality, functoriality, the Künneth formula, Poincaré duality, and Tate twists. The twist $\mathbb{Q}_\ell(r)$ is not merely a normalization convention: it is the precise correction required so that codimension-r geometry matches degree-$2r$ cohomology and so that Frobenius on a codimension-r algebraic class has weight 0 after twisting.

Proposition 2.1 (Basic structure)**.** *Let X/k be smooth projective of dimension d. Then:*

(i) $H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell) = 0$ *for $i < 0$ and $i > 2d$, and each group is finite-dimensional.*

(ii) *There is a perfect G_k-equivariant pairing*

$$H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell) \times H^{2d-i}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(d)) \longrightarrow \mathbb{Q}_\ell.$$

(iii) *For smooth projective X and Y there is a Künneth isomorphism*

$$H^n_{\text{ét}}((X \times_k Y)_{\bar{k}}, \mathbb{Q}_\ell) \cong \bigoplus_{i+j=n} H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell) \otimes H^j_{\text{ét}}(Y_{\bar{k}}, \mathbb{Q}_\ell).$$

(iv) *Any correspondence $\Gamma \in \mathrm{CH}^{d_X+r}(X \times Y)_{\mathbb{Q}}$ induces a graded map*

$$\Gamma_* \colon H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell) \longrightarrow H^{i+2r}_{\text{ét}}(Y_{\bar{k}}, \mathbb{Q}_\ell(r)).$$

The last point is the first place where the motivic shadow of the conjecture appears. Cohomology is not only a receptacle for classes; it is a representation of the ring of algebraic correspondences. The Tate conjecture will eventually ask whether all *arithmetically visible* endomorphisms of this representation come from that ring.

2.2. Algebraic cycles and equivalence relations

For a smooth projective variety X of dimension d, let $Z^r(X)$ denote the free abelian group on integral closed subschemes of codimension r. Passing to rational equivalence gives the Chow group $\mathrm{CH}^r(X)$. From the point of view of the Tate conjecture, the crucial fact is that Chow groups occupy only the first layer of a hierarchy. The deeper one moves toward cohomology, the weaker the equivalence relation becomes.

The basic chain is

$$\sim_{\text{rat}} \subseteq \sim_{\text{alg}} \subseteq \sim_{\text{hom},\ell} \subseteq \sim_{\text{num}} .$$

Rational equivalence is generated by principal divisors on codimension-$(r-1)$ subvarieties. Algebraic equivalence remembers how cycles move in connected algebraic families. Homological equivalence is defined by annihilation under a chosen realization, and numerical equivalence is the relation tested against all complementary cycles by the intersection pairing.

Definition 2.2. Let $\alpha \in Z^r(X)$.

(i) $\alpha \sim_{\text{alg}} 0$ if there exists a smooth connected curve T, a cycle $\mathcal{Z} \in Z^r(X \times T)$ flat over T, and points $t_0, t_1 \in T(k)$ such that $\alpha = \mathcal{Z}_{t_1} - \mathcal{Z}_{t_0}$.

(ii) $\alpha \sim_{\text{hom},\ell} 0$ if $\mathrm{cl}_\ell(\alpha) = 0$ in $H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))$.

(iii) $\alpha \sim_{\text{num}} 0$ if for every $\beta \in Z^{d-r}(X)$ one has $(\alpha \cdot \beta) = 0$.

A specialist will immediately recognize that the real tension lies in the gap between homological and numerical equivalence. Numerical equivalence is coarse enough to yield semisimplicity of the category of pure motives over a field, by Jannsen's theorem, whereas homological equivalence is the relation naturally seen by a realization functor. The standard conjecture of type D predicts that these two relations agree. In that sense the Tate conjecture is not only about the existence

of algebraic classes but also about the extent to which the realization-theoretic notion of triviality already coincides with the intersection-theoretic one.

There is another reason to keep the entire hierarchy in view. Codimension-one cycles are governed by the Picard scheme and thus admit a rich geometric structure; zero-cycles are tied to Albanese-type constructions; higher-codimension cycles have no equally simple parameter space. The increasing difficulty of the Tate conjecture in higher codimension reflects this geometric asymmetry.

A second point, often suppressed in brief expositions, is that the hierarchy of equivalence relations is already functorial in a distinctly algebro-geometric sense. Rational equivalence is local for the Zariski topology and is controlled by moving and deformation arguments; algebraic equivalence is measured by connected parameter spaces and hence interacts with the Picard and Albanese functors; numerical equivalence is detected by intersection theory and so belongs to the level of correspondences and bilinear forms. Thus the successive weakenings are not merely formal quotients: they successively forget the geometry of functions, then the geometry of families, and finally all information beyond intersection numbers. The Tate conjecture is difficult precisely because the ℓ-adic realization sees something intermediate between the family-theoretic and the intersection-theoretic levels.

For divisors this structure is especially transparent. The quotient of $\mathrm{Pic}(X)$ by algebraic equivalence is represented by the Néron–Severi group, while algebraically trivial line bundles form the connected group variety $\mathrm{Pic}^0_{X/k}$. In higher codimension there is no comparably satisfactory representability statement for $\mathrm{CH}^r(X)$ in general, and part of the force of the conjecture is that it predicts a finiteness and rigidity property for algebraic classes that the geometry of parameter spaces alone does not provide.

Remark 2.3. For later use it is convenient to regard a codimension-d_X cycle on $X \times X$ as a correspondence of degree 0. Composition is defined by pull-push along $X \times X \times X$, and it is only after passing to a sufficiently robust equivalence relation that idempotent projectors become manageable. This is the formal entry point of motives into the theory.

2.3. The cycle class map

The cycle class map is the precise point where intersection theory enters cohomology. Let $i\colon Z \hookrightarrow X$ be an integral closed subscheme of codimension r. If Z is smooth and regularly immersed, the purity isomorphism yields a local class in cohomology with supports, and pushing it forward to ordinary cohomology gives

$$\mathrm{cl}_\ell(Z) \in H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r)).$$

For general Z one proceeds by devissage: one first defines classes on the smooth locus, then extends across codimension-≥ 1 singularities using the localization sequence and compatibility with proper pushforward. The resulting map is additive on cycles and kills rational equivalences; hence it descends to

$$\mathrm{cl}_\ell\colon\ \mathrm{CH}^r(X) \otimes \mathbb{Q}_\ell \longrightarrow H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))^{G_k}.$$

It is important to stress that this map is not an auxiliary decoration. Its functorial properties

are exactly what allow one to compare the intersection-theoretic category of correspondences with tensor-linear operations on cohomology.

Proposition 2.4 (Functoriality package)**.** *The cycle class map is compatible with proper pushforward, flat pullback, refined Gysin maps for regular embeddings, cup products, and Chern classes.*

The compatibility with correspondences may be expressed by the following diagram. If $\Gamma \in \mathrm{CH}^{d_X+r}(X \times Y)$, then

$$\begin{tikzcd}
\mathrm{CH}^*(X \times Y) \otimes \mathbb{Q}_\ell \arrow[r, "\mathrm{cl}_\ell"] \arrow[d, "\text{composition of correspondences}"'] & H^{2*}_{\text{ét}}((X \times Y)_{\bar{k}}, \mathbb{Q}_\ell(*)) \arrow[d, "\text{convolution/cup-push}"] \\
\mathrm{Hom}\big(\mathrm{CH}^*(X), \mathrm{CH}^{*+r}(Y)\big) \arrow[r] & \mathrm{Hom}\big(H^\bullet_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell), H^{\bullet+2r}_{\text{ét}}(Y_{\bar{k}}, \mathbb{Q}_\ell(r))\big)
\end{tikzcd}$$

This diagram is intentionally schematic, but the message is exact: algebraic composition and cohomological composition are compatible. Thus every statement asserting that a cohomological morphism is algebraic is, at bottom, a statement about lifting a map across this square.

One should also keep in mind the codimension-one case of the construction. If L is a line bundle on X, then $\mathrm{cl}_\ell(c_1(L))$ coincides with the first ℓ-adic Chern class of L. The Tate conjecture for divisors is therefore the assertion that every invariant class of degree 2 and weight 0 is represented by a line bundle modulo algebraic equivalence. This is the arithmetic analogue of the Lefschetz $(1,1)$ theorem.

2.4. Supports, correspondences, and geometric coniveau

From the algebro-geometric side, the relevance of codimension-r classes is tied to the idea that a class should be visible on a closed subset of codimension at least r. The coniveau filtration packages this intuition by declaring that a cohomology class has coniveau at least r if it vanishes on the complement of some closed subset of codimension $\geq r$. Purity then shows that such classes arise from cohomology with supports, hence from Gysin images of classes on lower-dimensional subschemes. At a formal level this is the mechanism by which geometry feeds into the expected structure of the fixed part of cohomology.

This viewpoint is especially important for specialists because it clarifies what the cycle class map is trying to detect. A codimension-r cycle does not simply give a class in degree $2r$; more generally, a correspondence supported on a closed subset of large codimension constrains where the resulting endomorphism of cohomology can live with respect to coniveau and Lefschetz decompositions. In that sense the Tate conjecture is naturally adjacent to generalized Hodge-type questions: it predicts that the arithmetically distinguished pieces of cohomology already come from the part of the correspondence algebra with the correct support behavior.

2.5. Weil conjectures and Frobenius eigenspaces

When $k = \mathbb{F}_q$, the absolute Galois group is topologically generated by Frobenius. The action of geometric Frobenius on $H^i_{\text{ét}}(X_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ is therefore the arithmetic engine behind the Tate conjecture over finite fields.

Theorem 2.5 (Deligne)**.** *Let $X/\mathbb{F}_q$ be smooth projective. The eigenvalues of geometric Frobenius acting on $H^i_{\text{ét}}(X_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ are algebraic numbers all of whose complex absolute values equal $q^{i/2}$. The characteristic polynomial is independent of ℓ, and the action is semisimple.*

After twisting by (r), the fixed subspace is exactly the q^r-eigenspace in degree $2r$ before the twist. Thus for finite fields the Tate conjecture can be read as a precise claim about Frobenius eigenspaces.

Frobenius interpretation. For $X/\mathbb{F}_q$, the Tate conjecture in codimension r predicts that the eigenspace of Frob_q on $H^{2r}_{\text{ét}}(X_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ with eigenvalue q^r is spanned by algebraic classes.

3. Statement of the Tate conjecture and first reformulations

3.1. The basic statement

Conjecture 3.1 (Tate)**.** *Let k be finitely generated, X/k smooth projective, and $\ell \neq \mathrm{char}(k)$. Then for every integer $r \geq 0$ the cycle class map*

$$\mathrm{cl}_\ell \colon \mathrm{CH}^r(X) \otimes \mathbb{Q}_\ell \longrightarrow H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))^{G_k}$$

is surjective.

The conjecture is sometimes formulated with numerical equivalence already built in. This is not cosmetic: once numerical equivalence is used, the conjecture becomes a statement about correspondences and full faithfulness.

3.2. Easy cases and duality

The cases $r = 0$ and $r = \dim X$ are immediate. Poincaré duality also exchanges codimension r with codimension $d - r$. In particular the conjecture is symmetric under the transformation $r \leftrightarrow d - r$.

Proposition 3.2. *If X has dimension d, then the Tate conjecture in codimension r is equivalent to the corresponding dual statement in codimension $d - r$, provided one works with the natural perfect pairing coming from Poincaré duality.*

This symmetry is conceptually important: codimension-one classes are not an isolated phenomenon, but the first visible layer of a duality pattern that pervades the conjecture.

3.3. Divisors and the Néron–Severi group

For $r = 1$, algebraic cycles are divisors, and the conjecture acquires a much sharper geometric form. One has

$$\mathrm{CH}^1(X) = \mathrm{Pic}(X), \qquad \mathrm{NS}(X) = \mathrm{Pic}(X)/\,\mathrm{Pic}^0(X),$$

where $\mathrm{Pic}^0(X)$ is the connected component of the identity in the Picard scheme and $\mathrm{NS}(X)$ is the Néron–Severi group. The latter is finitely generated, and its rank $\rho(X)$ is the Picard number. The Tate conjecture in codimension one predicts that

$$\mathrm{NS}(X) \otimes \mathbb{Q}_\ell \xrightarrow{\sim} H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(1))^{G_k}.$$

What makes this case especially geometric is that both sides carry additional structure. On the left one has the intersection pairing, ampleness, the cone of divisors, and often lattice-theoretic information; on the right one has a Galois representation endowed with the cup-product form. The cycle class map is compatible with these structures, so the conjecture is not merely about dimensions but about the identification of two naturally polarized objects.

The Kummer sequence

$$0 \longrightarrow \mu_{\ell^n} \longrightarrow \mathbb{G}_m \xrightarrow{\ell^n} \mathbb{G}_m \longrightarrow 0$$

yields, after passing to cohomology and then to the inverse limit, an exact sequence that relates divisors, cohomology, and the Brauer group. In particular, modulo connected components and torsion issues, one is led to

$$0 \longrightarrow \mathrm{NS}(X_{\bar{k}}) \otimes \mathbb{Z}_\ell \longrightarrow H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Z}_\ell(1)) \longrightarrow T_\ell\,\mathrm{Br}(X_{\bar{k}}) \longrightarrow 0.$$

Thus the rational Tate conjecture for divisors may be viewed as the statement that the invariant subspace of degree-2 cohomology is exhausted by the Néron–Severi lattice, while integral refinements must confront the Brauer obstruction.

Remark 3.3. The divisor case is the arithmetic counterpart of the Lefschetz $(1,1)$ theorem not only formally but structurally. In both settings codimension-one algebraic classes are controlled by a geometric moduli object—the Picard scheme on the algebraic side, the Hodge decomposition on the complex side—and this extra structure is exactly what disappears in higher codimension.

4. Abelian varieties and the theorem of Tate

4.1. The Tate module and its Galois action

Let $A/\mathbb{F}_q$ be an abelian variety of dimension g. The ℓ-adic Tate module is

$$T_\ell(A) = \varprojlim_n A[\ell^n](\bar{\mathbb{F}}_q), \qquad V_\ell(A) = T_\ell(A) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

This is a $2g$-dimensional $\mathbb{Q}_\ell$-vector space equipped with a continuous action of $G_{\mathbb{F}_q}$. Cohomologically one has

$$H^1_{\text{ét}}(A_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell) \cong V_\ell(A)^\vee, \qquad H^i_{\text{ét}}(A_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell) \cong \bigwedge^i H^1_{\text{ét}}(A_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell).$$

4.2. Tate's isogeny theorem

Theorem 4.1 (Tate)**.** *For an abelian variety $A/\mathbb{F}_q$, the natural homomorphism*

$$\mathrm{End}_{\mathbb{F}_q}(A) \otimes \mathbb{Q}_\ell \longrightarrow \mathrm{End}_{G_{\mathbb{F}_q}}(V_\ell(A))$$

is an isomorphism.

This theorem is the model result behind the entire subject: it identifies geometric endomorphisms with Galois-equivariant linear algebra. The proof combines the semisimplicity of Frobenius, Honda–Tate theory, and a careful dimension count on central simple algebras attached to simple isogeny factors.

Proof strategy. The proof has four conceptual steps.

1. **Injectivity.** If an endomorphism acts trivially on $T_\ell(A)$, then it kills all ℓ^n-torsion and must vanish.
2. **Semisimplicity.** Frobenius acts semisimply on $V_\ell(A)$, so the right-hand side is the commutant of Frobenius.
3. **Honda–Tate classification.** The isogeny class of A is determined by the Frobenius polynomial. This transforms the geometry of A into the algebra of Weil numbers.
4. **Comparison of dimensions.** The endomorphism algebra of each simple factor is a central simple algebra over the field generated by Frobenius; the dimensions match the dimensions of the commutants on V_ℓ.

The injective map is therefore an isomorphism. □

The theorem immediately yields the Tate conjecture for divisors on abelian varieties over finite fields. Indeed, divisors can be encoded via homomorphisms to the dual abelian variety, and these in turn are detected on Tate modules.

Corollary 4.2. *Let $A/\mathbb{F}_q$ be an abelian variety. Then the Tate conjecture holds for A in codimension one. In fact, by exterior algebra and multiplicativity of cycle classes, it holds in all codimensions for powers of classes generated in degree one.*

4.3. Why abelian varieties are the first decisive case

Abelian varieties have extra structure: a group law, duality, polarizations, and an algebra of endomorphisms. This permits one to replace difficult cycle questions by endomorphism questions.

That mechanism is not available for general varieties, which is one reason the general Tate conjecture is so hard.

The geometry-to-linear-algebra passage may be visualized as

$$\begin{tikzcd}
\text{Divisors on } A \arrow[r] \arrow[d] & \mathrm{Hom}(A, A^\vee) \arrow[r] \arrow[d] & \mathrm{Hom}_{G_{\mathbb{F}_q}}(V_\ell(A), V_\ell(A^\vee)) \arrow[d] \\
\text{Cycle classes} \arrow[r] & \text{Correspondences} \arrow[r] & H^2_{\text{ét}}(A_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell(1))^{G_{\mathbb{F}_q}}
\end{tikzcd}$$

where the horizontal arrows encode, successively, line bundles, homomorphisms, and their realizations.

5. Abelian varieties over number fields and the theorem of Faltings

5.1. Why the characteristic-zero case looks different

Over a finite field the absolute Galois group is generated topologically by Frobenius, so the Tate conjecture becomes a problem about a single semisimple operator. Over a number field there is no global Frobenius endomorphism acting on cohomology. Instead one has a family of local Frobenius elements at places of good reduction and an ℓ-adic representation of a much larger Galois group. This is why the characteristic-zero theory requires a theorem of a different nature.

For abelian varieties the decisive substitute is Faltings's isogeny theorem. It identifies geometric homomorphisms with Galois-equivariant homomorphisms of Tate modules:

$$\mathrm{Hom}_k(A, B) \otimes \mathbb{Q}_\ell \xrightarrow{\sim} \mathrm{Hom}_{G_k}(V_\ell(A), V_\ell(B)).$$

When $A = B$ one recovers the endomorphism algebra. The theorem may be read as the characteristic-zero analogue of Tate's theorem over finite fields, but its proof is of a wholly different character: it uses the arithmetic of heights, reduction theory, and diophantine finiteness.

5.2. From endomorphisms to divisor classes

Let A/k be an abelian variety over a finitely generated field of characteristic 0. The divisor case of the Tate conjecture for A is obtained by combining Faltings's theorem with the geometry of the dual abelian variety $A^\vee$ and the relation between line bundles and homomorphisms.

A line bundle L on A determines a homomorphism

$$\varphi_L \colon A \longrightarrow A^\vee, \qquad x \longmapsto t_x^* L \otimes L^{-1},$$

and the assignment $L \mapsto \varphi_L$ factors through $\mathrm{NS}(A)$. After tensoring with $\mathbb{Q}$, the Rosati involution attached to a polarization identifies $\mathrm{NS}(A)_\mathbb{Q}$ with the symmetric endomorphisms inside $\mathrm{End}^0(A) := \mathrm{End}(A) \otimes \mathbb{Q}$. More precisely, if $\lambda \colon A \to A^\vee$ is a polarization, the Rosati

involution is

$$f \longmapsto f^\dagger := \lambda^{-1} \circ f^\vee \circ \lambda,$$

and one has an isomorphism

$$\mathrm{NS}(A)_{\mathbb{Q}} \cong \{f \in \mathrm{End}^0(A) \mid f^\dagger = f\}.$$

Applying V_ℓ and Faltings's theorem, one obtains

$$\mathrm{End}^0(A) \otimes \mathbb{Q}_\ell \cong \mathrm{End}_{G_k}(V_\ell(A)).$$

Passing to the Rosati-fixed part matches symmetric geometric endomorphisms with symmetric Galois-equivariant endomorphisms. Using the polarization to identify $V_\ell(A^\vee)$ with $V_\ell(A)(1)$, one is led to the degree-two cohomology group

$$H^2_{\text{ét}}(A_{\bar{k}}, \mathbb{Q}_\ell(1)) \cong \bigwedge^2 H^1_{\text{ét}}(A_{\bar{k}}, \mathbb{Q}_\ell)(1),$$

and the codimension-one Tate classes are exactly those arising from symmetric endomorphisms. Thus one recovers the Tate conjecture for divisors on abelian varieties over number fields.

Theorem 5.1 (Faltings, divisor case for abelian varieties)**.** *Let A be an abelian variety over a finitely generated field of characteristic* 0*. Then*

$$\mathrm{NS}(A) \otimes \mathbb{Q}_\ell \xrightarrow{\sim} H^2_{\text{ét}}(A_{\bar{k}}, \mathbb{Q}_\ell(1))^{G_k}.$$

Geometric explanation. The class of a line bundle is converted into a homomorphism $A \to A^\vee$; a polarization identifies these homomorphisms with Rosati-symmetric endomorphisms of A; Faltings identifies endomorphisms of A with Galois-equivariant endomorphisms of $V_\ell(A)$; and the symmetric part of the latter corresponds, through the cup-product pairing on H^1, to the invariant degree-two classes. The resulting chain of identifications is compatible with the first Chern class map, so the image of $\mathrm{NS}(A) \otimes \mathbb{Q}_\ell$ is exactly the Tate subspace. □

Remark 5.2. This argument illustrates a recurrent theme in the theory. The codimension-one case is tractable when divisors can be reinterpreted as endomorphisms or as linear data on a moduli object. Abelian varieties have this property because the Picard scheme is itself an abelian variety and the theorem of the square turns translation behavior of line bundles into homomorphisms.

6. Motivic reformulation and numerical motives

6.1. Pure motives and correspondences

Fix an adequate equivalence relation $\sim$ on cycles with rational coefficients. The category of pure motives $\mathsf{M}_\sim(k)$ is obtained by pseudo-abelian completion of the category whose objects are smooth projective varieties and whose morphisms are correspondences modulo $\sim$. Concretely,

$$\mathrm{Hom}_{\mathsf{M}_\sim(k)}\big(h(X), h(Y)(r)\big) = \mathrm{CH}^{d_X+r}(X \times Y)_{\mathbb{Q}}/\sim .$$

The relevance of this formula to the Tate conjecture is immediate: codimension-r cycles on X are morphisms from the unit motive to $h(X)(r)$.

The point of working motivically is that one can discuss projectors, decompositions, and endomorphism algebras before choosing a cohomology theory. If $p \in \mathrm{CH}^{d_X}(X \times X)_{\mathbb{Q}}/\sim$ is idempotent, then (X, p, m) is a motive and p cuts out a direct summand of every realization of X. In this language Künneth projectors become genuine geometric idempotents rather than merely cohomological decompositions.

6.2. Realization as a functor to representations

Any Weil cohomology theory yields a realization functor

$$R_\ell \colon \mathsf{M}_{\mathrm{hom}}(k) \longrightarrow \mathrm{Rep}_{\mathbb{Q}_\ell}(G_k),$$

and, after passing to numerical motives under suitable hypotheses, a realization

$$R_\ell \colon \mathsf{M}_{\mathrm{num}}(k) \longrightarrow \mathrm{Rep}_{\mathbb{Q}_\ell}(G_k).$$

On objects one has

$$R_\ell(h(X)) = H^\bullet_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell),$$

with grading and Tate twists understood; on morphisms the realization sends a correspondence to its induced map on cohomology.

The slogan that the Tate conjecture should be a full-faithfulness statement now becomes transparent. Evaluating the realization on the Hom-group from the unit motive to $h(X)(r)$ gives

$$\mathrm{Hom}_{\mathsf{M}_\sim(k)}(\mathbf{1}, h(X)(r)) \longrightarrow \mathrm{Hom}_{G_k}\Big(\mathbb{Q}_\ell, H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))\Big) = H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))^{G_k}.$$

But the source is precisely the group of codimension-r cycles modulo $\sim$. Thus the Tate conjecture is the assertion that this realization map is surjective when $\sim$ is chosen so as to retain algebraic correspondences while remaining compatible with the conjectural motivic formalism.

6.3. The moral content of full faithfulness

There are two distinct layers in the phrase "realization is fully faithful". Faithfulness says that a correspondence vanishing in all realizations should already be zero in the motivic category; fullness says that every cohomological morphism respecting the relevant symmetry should be induced by a correspondence. The Tate conjecture is the degree-one shadow of fullness, obtained by testing against the unit motive. Once one allows powers and summands of varieties, the same statement becomes a claim about all motivic morphisms.

Remark 6.1. This reformulation explains why numerical equivalence is so prominent. Jannsen's theorem shows that numerical motives form a semisimple abelian category. Semisimplicity is what allows one to pass from the algebraicity of classes to the algebraicity of projectors and, ultimately, to the control of all morphisms by a comparatively small set of generators.

6.4. Why semisimplicity matters algebraically

A decisive structural simplification occurs when one passes from homological to numerical equivalence. By Jannsen's theorem, the category of pure numerical motives over a field with rational coefficients is semisimple abelian. This fact is often quoted only as a categorical convenience, but it has a very concrete geometric consequence: idempotent correspondences split, nilpotent pathologies disappear, and arguments by decomposition become honest arguments inside the category rather than heuristic shadows of linear algebra.

In the present context semisimplicity matters because the Tate conjecture is expected to identify the commutant of arithmetic symmetries with an algebra of algebraic correspondences. Without semisimplicity, such commutants can carry extension data invisible on associated graded pieces; after passing to numerical motives, one may at least formulate the desired comparison in a category where orthogonal decompositions, isotypic pieces, and projector arguments behave as they do in representation theory. What remains conjectural is not the existence of the formalism but the claim that the realization functor still sees all morphisms one cares about.

The resulting tension can be summarized as follows. Homological equivalence is the relation adapted to a realization; numerical equivalence is the relation adapted to semisimplicity. The standard conjectures are precisely the bridge asserting that these two advantages occur simultaneously in the same category.

6.5. Hom-groups, weights, and the unit motive

It is useful to make the full-faithfulness slogan completely explicit. Let X and Y be smooth projective varieties of dimensions d_X and d_Y. In the category of pure motives modulo numerical equivalence one has

$$\mathrm{Hom}_{\mathsf{M}_{\mathrm{num}}(k)}\big(h(X)(r), h(Y)(s)\big) = \mathrm{CH}^{d_X+s-r}(X \times Y)_{\mathbb{Q}}/\mathrm{num}.$$

Thus every motivic morphism is represented by a correspondence of a very precise codimension. Once a Weil cohomology theory is chosen, such a correspondence acts degreewise on cohomology with the expected Tate twist. The question of fullness is therefore not vague: it asks whether every cohomological morphism compatible with the G_k-action and the tensor structure comes from a cycle of the prescribed codimension.

In particular, if one fixes a cohomological degree and passes to the appropriate Künneth summands, one is naturally led to comparison maps of the form

$$\mathrm{CH}^{d_X+s-r}(X \times Y)_{\mathbb{Q}}/\mathrm{num} \longrightarrow \mathrm{Hom}_{G_k}\big(H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell)(r), H^j_{\text{ét}}(Y_{\bar{k}}, \mathbb{Q}_\ell)(s)\big).$$

The role of algebraic Künneth projectors is precisely to isolate the (i, j)-piece geometrically rather than merely after realization. Without these projectors one can discuss the total action of a correspondence on $H^\bullet$, but not a refined motivic Hom-group attached to a single weight.

From this perspective the Tate conjecture is the first nontrivial test case of motivic fullness, namely the case in which the source is the unit motive and the target is a Tate twist of a single variety. Once the standard conjectures are granted, semisimplicity allows one to bootstrap from

this unit-motive test to arbitrary Hom-groups. In a semisimple rigid tensor category every object is built from summands of tensor constructions, and the algebraicity of the endomorphism algebra of a tensor generator controls the entire category. This is the structural reason the conjecture is so often phrased as a full-faithfulness problem rather than merely a statement about fixed vectors.

One can summarize the passage schematically as follows:

$$\begin{tikzcd}
\mathrm{Hom}_{\mathsf{M}_{\mathrm{num}}(k)}(\mathbf{1}, h(X)(r)) \arrow[r] \arrow[d, hook] & H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))^{G_k} \\
\mathrm{Hom}_{\mathsf{M}_{\mathrm{num}}(k)}\big(h(Y), h(X)(r)\big) \arrow[r] & \mathrm{Hom}_{G_k}\big(H^\bullet_{\text{ét}}(Y_{\bar{k}}, \mathbb{Q}_\ell), H^\bullet_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))\big)
\end{tikzcd}$$

The upper horizontal map is the classical Tate map. The lower horizontal map is its genuine motivic enlargement. The standard conjectures are what make the vertical passage from the top row to the bottom row geometrically meaningful rather than merely formal.

7. Künneth projectors, diagonals, and standard conjectures

7.1. The diagonal decomposition

Let X be smooth projective of dimension d. Cohomologically, the diagonal class

$$[\Delta_X] \in H^{2d}_{\text{ét}}(X \times X, \mathbb{Q}_\ell(d))$$

decomposes under Künneth as a sum of projectors

$$[\Delta_X] = \sum_{i=0}^{2d} \pi_i^{\mathrm{coh}}, \qquad \pi_i^{\mathrm{coh}} \in H^{2d-i}_{\text{ét}}(X, \mathbb{Q}_\ell) \otimes H^i_{\text{ét}}(X, \mathbb{Q}_\ell(d)).$$

These projectors cut out the graded pieces of cohomology. The algebro-geometric question is whether the π_i^{coh} are classes of algebraic correspondences on $X \times X$.

This problem is much more than a technical convenience. If the Künneth projectors are algebraic, cohomological decompositions lift to motivic decompositions. One can then speak of the piece $h^i(X)$ of the motive of X, not just of the ith cohomology group.

7.2. Standard conjectures and the Tate conjecture

The standard conjectures were designed precisely to bridge the gap between cohomological linear algebra and algebraic correspondences. Conjecture C predicts algebraicity of the Künneth projectors; conjecture D predicts equality of homological and numerical equivalence. Combined with the hard Lefschetz theorem and the expected algebraicity of the inverse Lefschetz operator, they provide the formal infrastructure needed to transport semisimple cohomological decompositions into the motivic category.

From the standpoint of the Tate conjecture, the importance of this package is twofold. First, it explains why numerical motives should control ℓ-adic realizations. Second, it isolates the precise obstruction to turning the algebraicity of fixed classes into full faithfulness of realization.

Proposition 7.1 (Motivic package, schematic form)**.** *Assume the standard conjectures in a form strong enough to guarantee algebraicity of Künneth projectors and the equality* $\hom = \mathrm{num}$ *on smooth projective varieties over the field under consideration. Then the semisimple category of numerical motives carries enough projectors for the Tate conjecture to be reformulated as the full faithfulness of ℓ-adic realization.*

7.3. Primitive decomposition and algebraicity of projectors

Fix a smooth projective variety X of dimension d together with an ample class $L \in \mathrm{CH}^1(X)_{\mathbb{Q}}$. On any Weil cohomology theory one has the hard Lefschetz isomorphisms

$$L^{d-i} \colon H^i(X) \xrightarrow{\sim} H^{2d-i}(X)(d-i),$$

and hence the primitive decomposition

$$H^i(X) = \bigoplus_{j \geq 0} L^j H^{i-2j}_{\mathrm{prim}}(X).$$

The standard conjectures predict that the inverse Lefschetz operators and the associated primitive projectors are induced by algebraic correspondences. This point is more rigid than the bare algebraicity of Künneth components. Once the primitive projectors are algebraic, the Lefschetz decomposition ceases to be purely cohomological: it becomes motivic.

For the Tate conjecture this matters in two ways. First, projectors isolate the piece of cohomology in which Tate classes can occur, removing the ambient noise coming from other degrees and Lefschetz powers. Second, algebraicity of primitive projectors allows one to transport statements about invariant classes into statements about direct summands of motives. This is one of the clearest points at which the conjecture ceases to be a problem about vector spaces and becomes a problem about the existence of algebraic idempotents.

7.4. A diagram for projectors and realizations

The relation among projectors, realizations, and correspondences may be compressed into the following square:

$$\begin{tikzcd}
\mathrm{CH}^d(X \times X)_{\mathbb{Q}}/\sim \arrow[r, "\mathrm{cl}_\ell"] \arrow[d, "\text{idempotents}"] & \mathrm{End}_{G_k}\big(H^\bullet_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell)\big) \arrow[d, "\text{cohomological projectors}"] \\
\text{Motivic summands of } h(X) \arrow[r] & \text{Graded } G_k\text{-stable summands of } H^\bullet_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell)
\end{tikzcd}$$

The upper horizontal arrow is the decisive one. The Tate conjecture concerns its surjectivity on the simplest morphisms; the standard conjectures are designed to guarantee that such surjectivity

is compatible with projector calculus. Together they explain why the conjecture is fundamentally about the reconstruction of algebraic structure from realization-theoretic data.

8. Coniveau, Lefschetz pencils, and support conditions

8.1. Coniveau as the geometric shadow of codimension

For a smooth projective variety X/k and an integer $r \geq 0$, the coniveau filtration on cohomology is defined by

$$N^r H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell) := \sum_{Z \subset X,\ \operatorname{codim} Z \geq r} \ker\Big(H^i_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell) \longrightarrow H^i_{\text{ét}}((X \setminus Z)_{\bar{k}}, \mathbb{Q}_\ell)\Big).$$

In other words, $N^r H^i$ consists of classes supported in codimension at least r. Every codimension-r algebraic cycle gives, through the local cycle class and the Gysin morphism, a class in $N^r H^{2r}(X_{\bar{k}}, \mathbb{Q}_\ell(r))$. This is a basic geometric fact, but it is worth isolating because it shows that the Tate conjecture is stronger than a mere invariance statement: it predicts that the distinguished G_k-fixed classes in degree $2r$ do not simply exist abstractly in cohomology, but in fact come from the deepest coniveau piece compatible with the degree.

The filtration is most useful when it is compared with the geometry of supports. If $j\colon U \hookrightarrow X$ is the complement of a closed subset Z of codimension at least r, then the long exact sequence with supports identifies the failure of a class to lie on U with the image of a Gysin map from Z. The coniveau language packages precisely the idea that a class should originate on a subvariety of controlled codimension. For cycle classes this is tautological; for arbitrary cohomology classes it is a genuine restriction.

8.2. Lefschetz pencils and the reduction of geometry to lower dimension

The coniveau filtration becomes geometrically effective when one uses Lefschetz pencils. Let $X \subset \mathbb{P}^N$ be smooth projective of dimension d, and choose a sufficiently general pencil of hyperplane sections. After blowing up the base locus one obtains a diagram

$$\begin{tikzcd} \widetilde{X} \arrow[r, "f"] \arrow[d, "\pi"] & \mathbb{P}^1 \\ X & \end{tikzcd}$$

whose generic fiber is a smooth hyperplane section of dimension $d-1$ and whose singular fibers have only ordinary quadratic singularities. The point of the construction is not only topological. In the algebro-geometric context it organizes codimension-r cycles on X relative to codimension-$(r-1)$ cycles on the hyperplane sections and to correspondences supported on the exceptional divisor.

On the cohomological side, the Leray spectral sequence for f and the vanishing-cycle formalism isolate the primitive part of cohomology, while the algebraic side keeps track of how cycles

restrict to the fibers and how their incidences vary in the family. This interaction is one of the classical sources of coniveau information: repeated hyperplane section arguments tend to force non-primitive pieces of cohomology into controlled support, leaving the primitive part as the true obstruction to algebraicity.

For the Tate conjecture, the moral is that many fixed classes should be testable fiberwise after one separates off the Lefschetz part. The decomposition

$$H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r)) = H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))_{\text{alg Lef}} \oplus H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))_{\text{prim}}$$

is not canonical in complete generality, but it captures the expected strategy: the non-primitive classes are generated by ample divisors and lower-dimensional algebraic data, whereas the primitive part encodes the genuinely new obstruction. This is exactly the pattern seen for surfaces, for cubic fourfold type questions, and for K3 surfaces, where the transcendental or primitive lattice is the hard core of the problem.

8.3. From support conditions to Tate classes

The relation between coniveau and Tate classes may be summarized schematically by the implication

$$\text{algebraic cycle of codimension } r \Longrightarrow N^r H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r)) \Longrightarrow H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))^{G_k}.$$

Only the first arrow is unconditional and geometric. The second arrow is not a theorem in this generality; it expresses the heuristic that Galois-fixed classes of the correct weight should arise from highly supported geometry. One may therefore regard the Tate conjecture as a far-reaching algebraization principle for the coniveau-r part of degree-$2r$ cohomology.

In practice, the coniveau viewpoint matters because it indicates where one should look for obstructions. If an invariant class cannot be shown to lie in the expected coniveau piece, then the problem is still transcendental from the algebro-geometric point of view. Conversely, whenever one can control the primitive part — by abelian methods, by Kuga–Satake, by decomposition of the diagonal, or by suitable standard conjectures — one often finds that the remaining classes are forced into algebraic support. This is why Lefschetz pencils and support filtrations continue to appear, even in highly arithmetic proofs, as a background geometric mechanism.

9. Crystalline methods and K3 surfaces

9.1. Why crystalline cohomology enters

When the base field has characteristic $p > 0$, ℓ-adic cohomology with $\ell \neq p$ does not see the full infinitesimal structure of the variety relative to p. For K3 surfaces this omission is serious: the deformation theory, the height of the formal Brauer group, and the slope decomposition of Frobenius are all inherently p-adic phenomena. Crystalline cohomology is therefore the natural replacement for de Rham cohomology in characteristic p, and the Frobenius endomorphism on crystalline cohomology refines the information carried by the ℓ-adic representation.

For a K3 surface X/k one considers the F-isocrystal

$$H^2_{\mathrm{cris}}(X/W(k)) \otimes_{W(k)} K,$$

equipped with the crystalline Frobenius. The Newton polygon of this F-isocrystal is controlled by the height of the formal Brauer group, while the Hodge polygon is determined by Hodge numbers. The interaction between these two polygons is one of the key geometric inputs in the analysis of the Tate conjecture for K3 surfaces.

9.2. K3 surfaces: Picard rank, formal Brauer group, and slopes

A K3 surface is a smooth projective surface X with trivial canonical bundle and $H^1(X, \mathcal{O}_X) = 0$. The Néron–Severi group $\mathrm{NS}(X_{\bar{k}})$ embeds into H^2 and has rank at most 22. The orthogonal complement of $\mathrm{NS}(X_{\bar{k}})$ inside second cohomology is the transcendental part; in characteristic p its behavior is reflected in the formal Brauer group $\widehat{\mathrm{Br}}(X)$.

The height $h(X) \in \{1, \ldots, 10, \infty\}$ of $\widehat{\mathrm{Br}}(X)$ governs the slope decomposition of the crystalline realization. Finite height corresponds to a mixed slope picture with a unique slope-1 part of rank $22 - 2h(X)$; supersingularity corresponds to the purely slope-1 case. From the perspective of the Tate conjecture this is decisive, because algebraic classes live in slope 1 (equivalently, after twist, in Frobenius slope 0), so the size of the slope-1 piece bounds the possible Picard rank.

In this form the geometry is already suggestive: the Tate conjecture for divisors on a K3 surface predicts that the entire slope-1 piece should be algebraic. The proof is difficult precisely because one must identify the whole relevant isocrystal piece with classes of line bundles rather than merely compare dimensions.

9.3. The Tate conjecture for K3 surfaces

For K3 surfaces over finite fields the divisor case of the Tate conjecture is now known. The original work of Artin–Swinnerton-Dyer and Nygaard–Ogus treated substantial classes of examples; the general case in odd characteristic was completed by Charles and Madapusi Pera, with the modern proof combining crystalline methods, deformation theory, and Kuga–Satake.

Theorem 9.1 (Tate for K3 surfaces over finite fields)**.** *Let $X/\mathbb{F}_q$ be a K3 surface and let $\ell \neq \mathrm{char}(\mathbb{F}_q)$. Then*

$$\mathrm{NS}(X_{\bar{\mathbb{F}}_q}) \otimes \mathbb{Q}_\ell \xrightarrow{\sim} H^2_{\text{ét}}(X_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell(1))^{G_{\mathbb{F}_q}}.$$

Structure of the proof. The argument has several layers.

1. One first separates finite-height and supersingular behavior through the crystalline slope decomposition.
2. For finite-height K3 surfaces, one uses the relation between the formal Brauer group and the slope filtration to control the relevant Frobenius-invariant subspace.
3. One then places the surface in a suitable moduli problem of polarized K3 surfaces and applies the Kuga–Satake construction to attach an abelian variety whose endomorphisms encode the primitive degree-two classes of the surface.

4. Finally one appeals to the Tate theorem for abelian varieties, together with a deformation argument, to show that the invariant classes come from line bundles.

The proof is not merely cohomological: it depends crucially on the algebraic geometry of moduli, quasi-polarizations, integral models of orthogonal Shimura varieties, and the control of special endomorphisms in families. □

9.4. Primitive cohomology and the transcendental part

Let X be a polarized K3 surface over a field and let $\xi \in \mathrm{Pic}(X)$ be the polarization class. The cup-product pairing on H^2 is nondegenerate, and the primitive cohomology is defined as the orthogonal complement of ξ with respect to this form. Over an algebraically closed field one may then separate the algebraic part generated by $\mathrm{NS}(X)$ from the remaining orthogonal complement. In characteristic zero this latter piece is the transcendental lattice; in positive characteristic one should be more careful with the available realizations, but the same pattern persists formally: the divisor classes occupy one distinguished summand, and the subtle part of the conjecture lies in showing that no extra fixed classes survive in the orthogonal complement.

This formulation is particularly useful because it isolates the genuine difficulty. For a K3 surface the middle cohomology is large, but the Tate conjecture for divisors says exactly that the Frobenius-invariant or Galois-invariant vectors in degree 2 are exhausted by the algebraic lattice. The orthogonal complement should therefore have no invariant vectors after the appropriate twist. In the crystalline proof, this is read off from slope considerations; in the Kuga–Satake approach, it is transferred to an endomorphism statement for an associated abelian variety.

9.5. A comparison diagram

The geometric input can be summarized by the schematic bridge

$$\begin{tikzcd}
\mathrm{NS}(X_{\bar{k}}) \otimes \mathbb{Q}_\ell \arrow[r] \arrow[d] & H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(1))^{G_k} \arrow[d] \\
\text{special endomorphisms of } \mathrm{KS}(X) \arrow[r] & \mathrm{End}_{G_k}\big(H^1_{\text{ét}}(\mathrm{KS}(X)_{\bar{k}}, \mathbb{Q}_\ell)\big).
\end{tikzcd}$$

The lower horizontal arrow is accessible by the theory of abelian varieties; the subtle issue is to show that the vertical arrows are sufficiently large and compatible with one another. This is exactly where the geometry of K3 moduli and the integral theory of Kuga–Satake enter.

10. The Kuga–Satake mechanism and orthogonal realizations

10.1. Primitive cohomology as an orthogonal object

For a polarized K3 surface (X, ξ), the primitive part of the second cohomology

$$P^2(X) := \xi^\perp \subset H^2(X)$$

carries a quadratic form coming from cup product. This orthogonal structure is the starting point of the Kuga–Satake construction. The basic idea is striking: instead of attacking the orthogonal representation $P^2(X)(1)$ directly, one embeds it into the endomorphism algebra of the first cohomology of a suitable abelian variety.

Theorem 10.1 (Kuga–Satake, schematic form)**.** *To a polarized complex K3 surface one can associate an abelian variety* $\mathrm{KS}(X)$ *such that the primitive Hodge structure of weight* 0 *on* $P^2_B(X)(1)$ *is realized inside the algebra*

$$\mathrm{End}\big(H^1_B(\mathrm{KS}(X),\mathbb{Q})\big)$$

through the spin representation of the even Clifford algebra.

The theorem is not merely a formal embedding. It transfers a difficult degree-two problem for a K3 surface to a degree-one problem for an abelian variety, where endomorphisms and moduli are much more accessible.

10.2. Why Kuga–Satake matters for the Tate conjecture

The arithmetic force of the Kuga–Satake construction is that it is compatible with realizations. In modern treatments one constructs integral models of the relevant orthogonal Shimura varieties and proves that the Kuga–Satake correspondence extends in families. This allows one to compare divisor classes on K3 surfaces with special endomorphisms of the associated abelian varieties.

$$\begin{tikzcd}
P^2_{\text{ét}}(X_{\bar{k}},\mathbb{Q}_\ell(1)) \arrow[r, hook] \arrow[d] & \mathrm{End}\big(H^1_{\text{ét}}(\mathrm{KS}(X)_{\bar{k}},\mathbb{Q}_\ell)\big) \arrow[d] \\
P^2_B(X_{\mathbb{C}},\mathbb{Q}(1)) \arrow[r, hook] & \mathrm{End}\big(H^1_B(\mathrm{KS}(X),\mathbb{Q})\big)
\end{tikzcd}$$

The vertical arrows represent comparison isomorphisms after choosing an embedding into $\mathbb{C}$ or passing through the appropriate p-adic comparison theory. The crucial point is that orthogonal invariants on the left are converted into endomorphism-type invariants on the right.

Conceptual payoff. The Kuga–Satake mechanism does not prove the Tate conjecture by itself. What it does is move the problem into a realm where Tate-style theorems for abelian varieties become available. In that sense it is a transport device from orthogonal geometry to linear-algebraic geometry.

10.3. Special endomorphisms and algebraicity

Once the Kuga–Satake variety is available, one studies the subspace of *special endomorphisms*: those endomorphisms of $\mathrm{KS}(X)$ whose realizations preserve the embedded primitive cohomology of the K3 surface. The key theorem in the modern approach is that these special endomorphisms are algebraic and that they recover the Tate classes of X in degree 2.

Proposition 10.2 (Bridge to Tate classes, schematic form)**.** *Let X be a polarized K3 surface*

over a finitely generated field of characteristic different from 2 *in a setting where the Kuga–Satake construction is available. Then the* G_k*-invariants in the primitive part of* $H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(1))$ *correspond to special endomorphisms of* $\mathrm{KS}(X)$.

This is exactly the point where one imports the theorems on endomorphisms of abelian varieties — Tate over finite fields and Faltings over number fields — back to the original K3 surface.

11. Integral refinements and the Brauer group

The rational Tate conjecture is only the first layer. Once one asks for integral statements, torsion enters, and the Brauer group becomes unavoidable.

For a smooth projective variety X/k, the Kummer sequence yields an exact sequence connecting the Picard group, ℓ-adic cohomology, and the ℓ-primary part of the Brauer group. In codimension one this is the source of the exact arithmetic correction term to an integral statement.

Proposition 11.1 (Kummer exact sequence)**.** *For* $\ell \neq \mathrm{char}(k)$ *there is an exact sequence*

$$0 \longrightarrow \mathrm{NS}(X_{\bar{k}}) \otimes \mathbb{Z}_\ell \longrightarrow H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Z}_\ell(1)) \longrightarrow T_\ell \operatorname{Br}(X_{\bar{k}}) \longrightarrow 0,$$

up to the usual finiteness caveats on torsion and connected components.

This explains why integral versions of the Tate conjecture are subtler: even when rational fixed classes are algebraic, the integral lattice can fail to be generated by cycle classes because a Brauer obstruction remains.

Remark 11.2. In codimension one, the obstruction to an integral statement is therefore not mysterious. It is measured by the ℓ-primary Tate module of the Brauer group. In higher codimension the integral picture becomes much more delicate and is related to unramified cohomology, torsion cycle classes, and refined cycle-theoretic invariants.

12. Tannakian interpretation and motivic Galois groups

12.1. Tannakian reconstruction

Suppose one is in a setting where numerical motives form a semisimple Tannakian category and R_ℓ is a fiber functor. Then one may define the motivic Galois group by

$$G_{\mathrm{mot}}(k, \ell) := \mathrm{Aut}^{\otimes}(R_\ell).$$

This group governs the tensor symmetries of all realizations simultaneously.

The guiding philosophy is that the classical Galois group G_k maps to $G_{\mathrm{mot}}(k, \ell)$, and the image should be Zariski dense. When the Tate conjecture holds, the image of Frobenius is already powerful enough to recover the relevant algebraic endomorphism algebras.

$$\begin{tikzcd} G_k \arrow[r, "\rho_\ell"] \arrow[dr, "\text{action on } H^\bullet"'] & G_{\mathrm{mot}}(k,\ell)(\mathbb{Q}_\ell) \arrow[d] \\ & \mathrm{Aut}^{\otimes}(R_\ell) \end{tikzcd}$$

12.2. Why the Tannakian viewpoint is useful

The Tannakian language packages a multitude of cohomological realizations into one symmetry object. In this setting, the Tate conjecture is no longer only about fixed vectors: it becomes the assertion that the category generated by realizations is controlled by algebraic correspondences, not by extra hidden linear algebra.

This is exactly the same phenomenon seen concretely in Tate's theorem for abelian varieties, now elevated to a categorical principle.

13. p-adic Hodge theory and comparison with Hodge classes

13.1. The characteristic-zero comparison problem

Over number fields one no longer has a single Frobenius element controlling the arithmetic. Instead one has a compatible system of local Galois representations together with comparison isomorphisms relating ℓ-adic, de Rham, and crystalline realizations at places above various primes. The natural question is then not only whether a class is G_k-invariant, but how that invariance is reflected in the Hodge filtration and in the local p-adic comparison data.

This is the environment in which the Tate conjecture should be compared with Hodge-type statements. A global Tate class is, in particular, de Rham at every place above ℓ and is constrained by the Hodge filtration after extension of scalars to $\mathbb{C}$. The arithmetic condition is therefore stronger than mere membership in a Hodge component, yet less visibly geometric. The difficulty is to recover algebraicity from this arithmetic rigidity.

13.2. Tate versus Hodge

For a smooth projective variety over $\mathbb{C}$, the Hodge conjecture predicts that rational (r,r)-classes in Betti cohomology are algebraic. The Tate conjecture is the arithmetic analogue obtained by replacing Hodge-theoretic symmetry by Galois-theoretic symmetry. The parallel is genuine but should not be overstated: Hodge classes are selected by a decomposition of a complex vector space, whereas Tate classes are selected by invariance under a profinite group action and must satisfy a web of local comparison conditions.

Dictionary.

Hodge side	Tate side
$H^{2r}(X(\mathbb{C}),\mathbb{Q}) \cap H^{r,r}$	$H^{2r}_{\text{ét}}(X_{\bar{k}},\mathbb{Q}_\ell(r))^{G_k}$
Lefschetz $(1,1)$ theorem	Tate for divisors
Mumford–Tate groups	ℓ-adic monodromy groups / motivic Galois groups
Absolute Hodge classes	expected compatible systems of Tate classes
Periods and Hodge filtrations	Frobenius, inertia, and comparison isomorphisms

A useful way to phrase the analogy is this: Hodge classes are the classes fixed by the Hodge cocharacter, while Tate classes are those fixed by the arithmetic Galois action. The conjectural bridge between the two is motivic. If an algebraic cycle is present, then it gives rise simultaneously to a Hodge class and to a compatible system of Tate classes. The remaining question is whether every class of the appropriate symmetry type arises in this way.

Remark 13.1. The codimension-one case again provides the cleanest comparison. The Lefschetz $(1,1)$ theorem says that integral $(1,1)$-classes are first Chern classes of line bundles. The Tate conjecture for divisors says that invariant degree-two classes with Tate twist are first Chern classes of line bundles. Beyond codimension one both conjectures become genuinely global and are no longer controlled by a moduli object as explicit as the Picard scheme.

14. Absolute Hodge classes, motivated classes, and compatible systems

14.1. From one complex realization to all realizations

Suppose now that X is defined over a field of characteristic 0 and that an embedding $k \hookrightarrow \mathbb{C}$ has been fixed. An algebraic cycle on X determines simultaneously a Betti class, a de Rham class, and for each prime ℓ an ℓ-adic Tate class whenever the cycle is defined over k. What is nontrivial is that these avatars are not unrelated: they are tied together by comparison isomorphisms and by the requirement that they behave functorially under correspondences, cup-products, and Künneth projectors. The resulting package is much more rigid than any single realization viewed in isolation.

This is the setting in which Deligne's notion of an *absolute Hodge class* enters. A Hodge class on a complex smooth projective variety is called absolute if it remains of Hodge type under every field automorphism of $\mathbb{C}$ and is stable under the full comparison package. The relevance of this notion to the Tate conjecture is conceptual rather than tautological: absolute Hodge classes are designed to isolate those Hodge classes that already behave as if they were motivic. For algebraic cycles this absoluteness is automatic. The conjectural content is that in favorable situations no genuinely new classes should appear.

14.2. Abelian varieties and the benchmark case

For abelian varieties, Deligne proved that all Hodge classes are absolute. This does not prove the Hodge conjecture for abelian varieties, but it does show that the Hodge-theoretic symmetry

type of such classes is stable under every realization. From the viewpoint of the Tate conjecture this is important because it suggests that the correct comparison object is not a single ℓ-adic invariant subspace, but an entire compatible system of realization-theoretic classes. Faltings's theorem then identifies the degree-one and divisor-type pieces of these systems with actual endomorphisms and symmetric homomorphisms.

The philosophical gain is substantial. Once one stops treating each ℓ separately and instead asks for a class that persists across all realizations, the expected algebraicity becomes more believable. A random fixed vector in one ℓ-adic representation may well have no geometric meaning; a class that is simultaneously Betti, de Rham, crystalline, and ℓ-adic in a compatible way is already constrained by the full motivic formalism.

14.3. André's motivated classes and the intermediate bridge

A useful intermediate notion is provided by André's category of motivated cycles. Here one enlarges the ring of algebraic correspondences just enough to obtain a semisimple Tannakian category without assuming the standard conjectures. The price is that motivated classes need not be known to be algebraic; the benefit is that one can nevertheless discuss projectors, tannakian groups, and realization functors in a robust unconditional framework.

For the present exposition, the importance of motivated classes is not technical but structural. They offer a language in which one can say: the standard conjectures would identify motivated, absolute Hodge, and algebraic classes; until that identification is known, motivated classes serve as a placeholder for the conjectural geometric category. Thus, in specialist terms, the Tate conjecture may be read as the statement that the G_k-invariant part of the relevant ℓ-adic realization contains no classes beyond the algebraic subspace, even though one may first encounter it naturally inside a larger motivated or absolute-Hodge envelope.

14.4. Compatible systems and the real content of independence of ℓ

The phrase "independence of ℓ" is sometimes used too loosely. What is really at stake is not merely the constancy of dimensions, but the existence of a realization-theoretic packet whose members correspond under all comparison isomorphisms and whose symmetry groups have the expected common origin. For algebraic cycles this packet is produced functorially. For conjectural Tate classes one expects the converse: if a class appears as part of a compatible family with the correct Hodge and de Rham constraints, then it should be motivic and hence algebraic.

This viewpoint leads to a more refined narrative for the Tate conjecture. One does not ask whether a lone class in $H^{2r}_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(r))^{G_k}$ is algebraic. One asks whether the entire compatible system to which it ought to belong can be generated by algebraic correspondences. Put differently, the conjecture is not fundamentally about one fixed vector space but about the coincidence of two tannakian envelopes: the one generated by realizations and the one generated by cycles.

15. Open problems

Despite dramatic progress, the general Tate conjecture remains open. The main unsolved territories include higher codimension for general varieties, K3 surfaces over number fields, and systematic integral refinements.

15.1. Representative open problems

1. **Higher codimension over number fields.** Even for abelian varieties, the conjecture in codimension $r > 1$ is largely open in general.
2. **K3 surfaces over number fields.** The divisor case over finite fields is solved, but the corresponding arithmetic statement over number fields remains subtle.
3. **Algebraicity of Künneth projectors.** This is central to any motivic decomposition of cohomology.
4. **Independence of ℓ.** The dimension of the Tate subspace is expected to be independent of ℓ in broad generality, but this is far from settled in all settings.
5. **Integral forms.** Rational surjectivity is only the beginning; torsion and Brauer phenomena complicate integral statements.

15.2. Why progress is difficult

Each known proof uses extra geometry.

- Abelian varieties admit group structures and a rich endomorphism algebra.
- Products of curves reduce to linear algebra built from Jacobians.
- K3 surfaces rely on deformation theory, crystalline methods, and Kuga–Satake.

A general theory would need a way to detect algebraicity directly from cohomological symmetry, without the presence of special structures. This is precisely where the conjectural motivic framework becomes indispensable.

16. Appendix A: Weil cohomology and correspondences

A Weil cohomology theory on smooth projective varieties over a field k with coefficients in a characteristic-zero field F is a contravariant functor $H^\bullet$ satisfying finite-dimensionality, Künneth, Poincaré duality, cycle classes, and a Lefschetz-type package. Examples include Betti cohomology over $\mathbb{C}$, de Rham cohomology in characteristic zero, and ℓ-adic cohomology for $\ell \neq \mathrm{char}(k)$.

Correspondences act on any Weil cohomology theory. If $\Gamma \in \mathrm{CH}^{\dim X + r}(X \times Y)$, then Γ induces a map

$$\Gamma_* \colon H^i(X) \longrightarrow H^{i+2r}(Y)(r).$$

This is the formal mechanism through which the category of motives interacts with cohomology. The Tate conjecture concerns the extent to which every cohomologically natural map of this form comes from an actual algebraic correspondence.

17. Appendix B: The Lefschetz trace formula and counting points

For $X/\mathbb{F}_q$ smooth projective, the zeta function

$$Z(X/\mathbb{F}_q, T) = \exp\left(\sum_{n\geq 1} \frac{\#X(\mathbb{F}_{q^n})}{n} T^n\right)$$

admits the cohomological factorization

$$Z(X/\mathbb{F}_q, T) = \prod_{i=0}^{2d} \det\big(1 - T\,\mathrm{Frob}_q \mid H^i_{\text{ét}}(X_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell)\big)^{(-1)^{i+1}}.$$

This is the most concrete bridge between arithmetic and cohomology. Counting points gives traces of Frobenius; the Tate conjecture interprets special eigenspaces of Frobenius as coming from algebraic geometry in codimension r.

18. Appendix C: A longer motivic package

It is useful to isolate the network of conjectural and proven implications that repeatedly reappears throughout the subject. The Tate conjecture is rarely used in isolation; instead it sits inside a larger package involving semisimplicity, projectors, and comparison between equivalence relations.

$$\begin{tikzcd}
\text{Standard conjectures} \arrow[r] \arrow[d] & \text{Algebraic Künneth projectors} \arrow[d] \\
\text{hom} = \text{num on cycles} \arrow[r] & \text{Semisimple category of numerical motives}
\end{tikzcd}$$

Once these ingredients are in place, realization functors become much better behaved. In particular, one expects the ℓ-adic realization to detect morphisms faithfully and to identify motivic endomorphisms with Galois-equivariant endomorphisms of cohomology. This is the precise sense in which the Tate conjecture becomes a statement about full faithfulness.

Indeed, for smooth projective X and Y one would like the realization functor to induce an isomorphism

$$\mathrm{Hom}_{\mathsf{M}_{\mathrm{num}}(k)}\big(h(X), h(Y)(r)\big) \otimes \mathbb{Q}_\ell \xrightarrow{\sim} \mathrm{Hom}_{G_k}\big(H^\bullet_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell), H^{\bullet+2r}_{\text{ét}}(Y_{\bar{k}}, \mathbb{Q}_\ell(r))\big).$$

When $X = \operatorname{Spec} k$ this reduces exactly to the cycle class map onto invariants. The point is that the conjecture should hold uniformly for all powers, summands, and correspondences, not only

for classes on a single variety.

Proposition 18.1 (Motivic consequence, schematic form)**.** *Assume enough of the standard conjectural package so that pure numerical motives over a finite field form a semisimple Tannakian category and the relevant Künneth projectors are algebraic. Then the Tate conjecture is equivalent to the full faithfulness of the ℓ-adic realization on pure motives.*

Explanation. Full faithfulness means that every Galois-equivariant cohomological morphism between realizations is induced by an algebraic correspondence. Evaluating this statement on the unit motive and a Tate twist of $h(X)$ recovers exactly the surjectivity of the cycle class map onto the invariant part. Conversely, if Tate classes are algebraic in all powers and if the cohomological projectors are algebraic, then the decomposition of morphisms into Künneth components lifts from cohomology to correspondences. Semisimplicity of numerical motives then promotes surjectivity on distinguished generators to fullness on all morphism spaces. □

Remark 18.2. This appendix explains why the Tate conjecture is so often paired with the slogan that motives should be reconstructed from realizations. The conjecture is not merely about a vector subspace of cohomology; it is about whether realization theory forgets any genuinely algebraic morphisms. In this respect the passage from classes to correspondences is not optional but intrinsic to the problem.

19. Conclusion

The Tate conjecture is best understood as a statement about the passage from geometry to realization. In its simplest form it asks whether the fixed part of ℓ-adic cohomology is generated by algebraic cycles. In its more structural form it asks whether the Galois-equivariant linear algebra attached to a smooth projective variety is already controlled by the algebra of algebraic correspondences. The second formulation contains the first and explains its persistence across such different geometric settings.

From this point of view, the recurrent themes of these notes are not accidental. Absolute purity explains why codimension enters with the twist that it does. The Picard scheme and Néron–Severi lattice show how the conjecture becomes concrete in codimension one. Semisimplicity of numerical motives and algebraicity of Künneth projectors explain why full faithfulness is the correct motivic reformulation. The theory of abelian varieties exhibits the conjecture in a setting where endomorphism algebras can actually be computed, while the case of K3 surfaces shows how much more delicate the geometry becomes once the middle cohomology contains a large primitive part.

What remains open is therefore quite precise. One must understand whether the cohomological symmetries singled out by arithmetic admit an algebraic origin uniformly, compatibly, and in all codimensions. That question is local in appearance but global in substance: it sits at the meeting point of intersection theory, the geometry of parameter spaces, comparison isomorphisms, and the conjectural structure of pure motives.

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